

Environmental, Water & Biodiversity Screening

Environmental approvals, habitat constraints, protected species, wetlands, drainage, water use, and mitigation.

Project: Solar Alpha **Location:** Solano County, California (fictional demo site) **Date:** August 14, 2026

Demo note: This document contains intentionally synthetic data and seeded inconsistencies for a one-day hackathon prototype. It is not a real engineering, legal, environmental, investment, or regulatory opinion.

Screening summary

Desktop and field screening identified seasonal habitat, drainage features, and a protected raptor nesting area within the broader study boundary. No project-wide exclusion is recommended, but several features require avoidance buffers or agency consultation.

Topic	Finding	Status
Wetlands / waters	4.6 acres potentially jurisdictional	Field delineation pending
Seasonal drainage	9 mapped swales	Avoid / bridge where feasible
Protected raptor habitat	2 active nesting territories	Consultation required
Special-status plant survey	No observations in spring pass	Second seasonal pass pending
Construction water demand	82 acre-feet total	Source not fully contracted
Dust control	High summer demand	Mitigation plan required

Biodiversity findings

Two active protected-raptor nesting territories were documented during the April survey. The preliminary layout places approximately 46 acres of panels inside the recommended seasonal disturbance buffer.

RED FLAG - The current layout conflicts with the preliminary nesting buffer. Re-layout could reduce DC capacity by 8-12 MW unless the schedule avoids the sensitive season or consultation supports a modified buffer.

Water flows and drainage

Hydrology mapping indicates concentrated stormwater flow through the central and southeast blocks. The 30% civil design assumes three crossings. The land report references nine drainage corridors, which is consistent with the desktop map but broader than the current civil design basis.

WARNING - Civil design has not yet incorporated all mapped drainage corridors. Additional culverts or setbacks may be required.

Construction water supply

The construction plan estimates 82 acre-feet for dust suppression and grading. A letter from the local water district expresses willingness to discuss up to 50 acre-feet, subject to annual allocation and drought restrictions.

RED FLAG - 32 acre-feet of projected construction water demand is currently uncontracted.

Environmental approval status

Approval / study	Planned completion	Dependency
Wetland delineation	Oct 2026	Final grading plan
Raptor consultation	Dec 2026	Construction timing / layout
Stormwater plan	Jan 2027	Civil IFC package
Construction water agreement	Nov 2026	Mobilization

These dates should be cross-checked against the regulatory schedule and financial model for sequencing conflicts.